

When Code and Oracle Disagree: A Receipt-Gated Feasibility Study for Multi-Agent Adjudication

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Abstract

Agentic evaluations often plan from benchmark or artifact counts, but primary analyses may require compound units that can disappear across construction, qualification, review, and constrained assembly. We introduce Receipt-Gated Compound-Unit Accounting, a method that uses typed analysis units aligned to the estimand, records every gate transition, solves the constrained assembly of eligible primary analysis units, and binds progression to a pre-specified adequacy calculation. We apply the method to a planned matched-depth, four-invocation comparison of two multi-agent code adjudication protocols on role-swapped program pairs. From 114 source tasks, construction produced 228 programs. Qualification and review left 101 clean and 69 fault programs, while task identity and lineage requirements reduced this supply to 15 mirror-capable tasks. The historical total-item-first allocation contained eight pre-gold mirror candidates. A unit-first re-audit found 14 under the same pilot quotas and 15 when supplemental minima were removed. Even the 15-task identity ceiling provided only 0.098 power under the locked analysis. Applying the typed audit to the preserved receipt trail exposed an earlier unit-accounting error that treated all 51 selected items as paired units. Correcting the denominator to mirror tasks triggered a stop before final-gold adjudication and before any main comparison call. At eight historical candidates, achieved power under the pre-specified alternative with $\delta=0.2$ was 0.016. A separately implemented exact enumeration reproduced all reported values and found that the original 60-pair resource target provided only 0.525 power, while 103 pairs were required for 0.8 power. We preserve the original target receipt and record this correction in a successor adequacy receipt. Our contributions are a typed gate model, a constrained assembly formulation, an adequacy gate, and a worked validation case in retrospective compound-unit and power-plan auditing. We make no protocol-performance claim.

Keywords

multi-agent systems, agentic software engineering, LLM code review, fault attribution, test oracle problem

1 Introduction

Agentic evaluations often plan from benchmark or artifact counts, but primary analyses may require compound units. A matched comparison may require several independent artifacts, a shared task identity, a qualifying fault, a complete role swap, and an admissible gold label before one observation exists. A large program pool can therefore support few statistical units. Power calculated from the pool size answers the wrong question.

Multi-agent evaluations are particularly exposed to compound-unit attrition because one benchmark identity can expand into arm-specific sessions, role-specific views, paired presentations, shared execution evidence, and a sealed gold join. Agent calls and

generated programs are therefore not interchangeable with the task-level units consumed by the comparative estimand.

The distinction matters when two programs disagree. The candidate alone may be faulty. The designated oracle alone may be faulty. Both programs may be faulty, neither may be faulty, or the specification may be materially ambiguous. Execution establishes divergence without deciding among these five judgments. Expanded tests reveal behavior that smaller suites miss [12]. Mutation and differential testing make differences reproducible [6, 13]. The testing oracle problem remains because reproducible output can still conflict with intended behavior [1].

We planned a matched-depth comparison of repeated full-view review against routed specialist review. Separately constructed HumanEval+ lineages would form role-swapped mirror pairs. A pair would become one eligible primary analysis unit only after both orientations had eligible programs and final adjudication. Preparation yielded 101 qualified clean programs and 69 qualified fault programs. Task identity reduced this supply to 15 mirror-capable tasks. The locked total-item-first selection returned eight pre-gold mirror-task candidates. Final adjudication could only reduce that count within the selected dataset.

The power plan also contained a mathematical error. Under the stated paired risk difference and discordance, the 60-pair original resource target did not provide the intended 0.8 power. A successor adequacy receipt records the correction while preserving the historical target receipt. This is a methodology validation case. The gate sequence and no-call boundary were prospective. The typed formalization, selector re-audit, and power correction were retrospective methodological analyses of the preserved trail.

This paper contributes the following.

- Receipt-Gated Compound-Unit Accounting types the analysis unit emitted by every gate and records exclusions, unresolved cases, and infrastructure failures without silently changing denominators.
- A constrained assembly procedure constructs eligible primary analysis units under lineage, fault, task-disjointness, reservation, separation, and control-quota requirements. An adequacy function then binds proceed, redesign, or stop to the locked analysis.
- A preserved validation case shows why unit typing matters. A superseded receipt counted all 51 selected items as pairs. The successor receipt uses the eight pre-gold mirror-task candidates in the executed selection and records stop.
- The executed case study traces compound-unit attrition from program construction through selection. It retains the planned G3R and M4 protocols as the motivating design while making no claim about their performance.

2 Related Work

2.1 Multi-agent evaluation and controlled depth

ChatDev, MetaGPT, and AutoGen organize language-model calls into roles and conversations for software work [5, 15, 19]. Debate and independent sampling offer other paths from several judgments to one answer [3, 11]. Their results depend on orchestration as well as agent count. Equal thinking-token budgets can remove an apparent multi-agent advantage [18]. Trace studies locate failures in system design, inter-agent alignment, and task verification [2]. MAS-Orchestra treats orchestration choices as the experimental object [8].

Budget-controlled evaluation motivates the design. The planned protocols match call count and depth, while token use and latency remain measured rather than assumed equal. G3R and M4 each specify three parallel upstream calls, one shared deterministic executor, and one resolver. They use the same requested model and execution limits. The planned contrast concerns the complete protocols because routed views, specialist roles, and evidence flow change together.

2.2 Feasibility and progression criteria

Pilot and feasibility methodology distinguishes testing whether a definitive study can be done from testing the definitive effectiveness hypothesis. The CONSORT extension for randomized pilot and feasibility trials asks authors to state progression criteria and report how pilot evidence informs the future study [4]. The present setting is not a clinical trial. The relevant methodological principle is the separation of feasibility from a confirmatory claim. A locked progression rule should be evaluated on the unit used by the future analysis.

Software and agent evaluations often report substrate yield without an explicit adequacy gate. Our method connects progression criteria to typed analysis units aligned to the estimand. An underpowered assembly cannot be relabeled as a smaller confirmatory study after its outputs are known.

Estimand frameworks emphasize aligning study design, data collection, and analysis with the exact target quantity [7]. Our concern is the upstream construction and accounting of the compound analysis units required to estimate that quantity.

2.3 Workflow provenance

in-toto records ordered supply-chain steps and verifies the materials and products associated with them [17]. RO-Crate packages research artifacts with structured metadata [16]. Workflow Run RO-Crate captures inputs, outputs, code, and execution context across workflow systems [10]. These systems motivate durable descriptions of what a stage consumed and produced. We do not claim content addressing or provenance as novel.

Provenance records what was consumed and produced. Our method additionally types the statistical unit each stage produces, solves the constrained assembly of eligible primary analysis units, recomputes the locked analysis, and binds stop and deviation decisions to the adequacy result. The novelty lies in this analysis-unit accounting layer rather than in the receipt substrate.

3 Receipt-Gated Compound-Unit Accounting

3.1 Typed study units

Let each stage operate on an explicit unit universe. The units used here are

$\mathcal{U}_T$	task identities,
$\mathcal{U}_K$	construction packets,
$\mathcal{U}_P$	accepted programs,
$\mathcal{U}_F$	frozen mutation candidates,
$\mathcal{U}_W$	witnessed fault candidates,
$\mathcal{U}_R$	reviewed programs,
$\mathcal{U}_I$	independently checked programs,
$\mathcal{U}_{QC}$	qualified clean programs,
$\mathcal{U}_{QF}$	qualified fault programs,
$\mathcal{U}_{MC}$	mirror-capable tasks,
$\mathcal{U}_M$	role-swapped mirror pairs,
$\mathcal{U}_S$	supplemental items,
$\mathcal{U}_A$	eligible primary analysis units.

A task is a stable benchmark identity. A construction packet is one isolated request for one lineage of one task. A program is accepted source from that packet. A frozen mutation candidate is selected without outcome access from an outcome-free ranked universe. A witnessed fault candidate carries the required reproducible divergence evidence. A reviewed program carries a completed panel verdict. An independently checked program carries a fresh-input check. A qualified clean program has passed the clean path. A qualified fault program has passed the fault, witness, review, and independent evidence requirements. A mirror-capable task has two independently constructed clean lineages and a qualifying fault in the required lineage. A mirror pair is the two role-swapped presentations assembled from those programs. A supplemental item belongs to a locked control or challenge stratum. An eligible primary analysis unit is a mirror pair that also survives every later eligibility requirement.

The distinctions prevent conversion by rhetoric. Two programs do not automatically form a task. Two items do not automatically form a pair. A pre-gold mirror candidate does not automatically become an analyzable primary unit. Counts may decrease as evidence accumulates but cannot increase merely because a downstream analysis needs a larger denominator.

3.2 Typed gates and receipts

For stage s , a gate is an evidence-producing map over stage jobs

$$g_s : \mathcal{J}_s \longrightarrow \mathcal{U}_s \uplus \{\text{EXCLUDED, UNRESOLVED, INFRASTRUCTURE FAILURE}\}. \quad (1)$$

Each stage job $j \in \mathcal{J}_s$ references a finite predecessor set $P(j) \subseteq \bigcup_r \mathcal{U}_r$. A stage job may bind one or more predecessor units. Multiple jobs may reference the same predecessor, allowing branching, while a downstream job may bind several predecessors. Constrained dataset assembly remains a separate many-to-many operation.

The disjoint result states keep scientific exclusion separate from missing judgment and failed execution. A retry policy may resolve an infrastructure failure. It does not turn that state into an eligible unit without a recorded successful successor.

Every gate receipt records the stage-job identity, input unit types, output unit type, predecessor hashes, eligibility decision, reason, attempt status, and successor identity. The successor identity names the next unit when one exists. It also prevents a program, item, and pair from sharing one ambiguous identifier. Receipts are append-only study records. A correction writes a successor receipt that names the superseded record and preserves its bytes.

This contract supports local verification. It does not prove that an operator who controls the evidence root could not delete or replace the entire root. Without an external anchor, receipts are not tamper evidence. Hashes establish equality to preserved bytes and no more.

3.3 Constrained assembly

Let binary x_t indicate that task t supplies a mirror candidate. Let binary z_{tk} indicate assignment of task t to supplemental stratum k . The binary constants c_{t1} and c_{t2} mark qualified clean programs from independent construction packets. The constants f_t and e_{tk} mark the required qualified fault and supplemental eligibility. Let R_D contain development identities reserved outside the main assembly. Let R_C contain the locked challenge reservation pool and let $F_C \subset R_C$ contain the challenge identities selected as a fixed main stratum. Variable mirror and supplemental assignments exclude $R_D \cup R_C$.

Let $\mathcal{I}(x, z)$ be the main-item set containing two presentations for each selected mirror task, one item for each selected supplemental task, and the fixed challenge items indexed by F_C . A deterministic schedule $\pi: \mathcal{I}(x, z) \rightarrow \{1, \dots, |\mathcal{I}(x, z)|\}$ is a bijection. The function H maps the canonical assignment and schedule to its nonnegative integer rank under the locked hash rule.

For a declared supplemental case, the selector first verifies that every active minimum stratum is satisfiable and restricts the domain to assignments that meet those minima. On that admissible domain it uses the lexicographic objective

$$\text{lexmax}_{x, z, \pi} \left(\sum_t x_t, \sum_{t, k} z_{tk}, -H(x, z, \pi) \right), \quad (2)$$

where L_k is the minimum for stratum k . The operational priorities are to maximize primary mirror tasks, require every active supplemental minimum, maximize supplemental items, and break remaining ties deterministically. Minimum satisfaction is an eligibility condition, not an objective term that becomes constant on the admissible domain.

The core constraints are

Algorithm 1. Receipt-gated compound-unit accounting

- (1) Declare the analysis unit required by the primary estimand and every predecessor unit type.
- (2) Apply each gate job to a verified predecessor set and append one typed receipt for every eligible, excluded, unresolved, or failed result.
- (3) Assemble candidate primary and supplemental units under the locked constraints using Eq. 2.
- (4) Set N^* to the verified candidate upper bound or to the final eligible count when adjudication is complete.
- (5) Recompute the locked analysis with N^* . When monotonicity is not established, maximize adequacy over every attainable count up to N^* . Then evaluate the adequacy function.
- (6) Write the progression decision. A correction names and preserves its predecessor in a successor receipt.
- (7) Release protected adjudication or main calls only after PROCEED. A redesign receives a new protocol identity. A stop retains the no-call boundary.

$$x_t \leq c_{t1}, \quad x_t \leq c_{t2}, \quad x_t \leq f_t, \quad (3)$$

$$z_{tk} \leq e_{tk}, \quad x_t + \sum_k z_{tk} \leq 1, \quad (4)$$

$$x_t = z_{tk} = 0 \quad \text{for } t \in R_D \cup R_C, \quad (5)$$

$$L_k \leq \sum_t z_{tk} \leq U_k \quad \text{for each active } k, \quad (6)$$

$$x_t = 1 \implies |\pi(m_t^1) - \pi(m_t^2)| > 1. \quad (7)$$

The two clean flags in Eq. 3 refer to different packet hashes and therefore different construction lineages. The fault flag includes the locked parent and witness requirements. Equation 4 keeps a task out of multiple strata. Equations 5 and 6 enforce development exclusion, challenge-pool reservation, and variable control quotas. The fixed challenge quota is $|F_C|$ and is checked before optimization. In the case study, $|R_C| = 10$ and $|F_C| = 8$. Equation 7 prevents adjacent mirror mates. Schedule bijectivity prevents position reuse. Further resource constraints can prevent reuse of a physical program across items.

3.4 Adequacy and progression

Let N^* be the assembled upper bound or the final count of eligible primary analysis units. Let δ be the pre-specified paired risk difference, d the discordance rate, α the two-sided test level, and $1 - \beta$ the target power. The adequacy function is

$$A(N^*, \delta, d, \alpha, 1 - \beta) \longrightarrow \{\text{PROCEED}, \text{REDESIGN}, \text{STOP}\}. \quad (8)$$

PROCEED requires the locked power calculation to meet $1 - \beta$ and every non-power gate to pass. REDESIGN applies only before protected main outputs exist and requires a new protocol identity for a changed substrate, estimand, or analysis. STOP applies when the locked study cannot meet adequacy and no authorized redesign is entered. Neither redesign nor stop authorizes an underpowered primary comparison.

An upper bound is decisive when the locked adequacy measure is monotone in the eligible-unit count over the attainable range. Otherwise, the gate evaluates the maximum adequacy over every attainable $n \leq N^*$. In this case, exact enumeration confirmed inadequacy for all $n \leq 15$. Final adjudication can only remove units, so the call boundary remains closed.

4 Case Study Design

4.1 Five-way adjudication

The motivating case asks which source is responsible when a candidate and a designated oracle disagree. Final decisions allow candidate-only fault, oracle-only fault, faults in both, neither faulty, or material specification ambiguity. A resolver may abstain. Transport, timeout, schema, and client failures are execution states rather than semantic labels. Material ambiguity applies only when plausible readings change program correctness and prevent a unique attribution.

The planned final-gold process would record specification status and the status of neutral Program A and Program B. Two reviewers would judge each packet while blinded to system outputs, program provenance, candidate and oracle roles, mutation history, mirror links, and the other reviewer’s answer. Only axis disagreements would reach a third reviewer. This stage was not entered.

4.2 Matched-depth, four-invocation protocols

G3R assigns three independent generalists the specification and both role-labelled programs. A strict plurality supplies a diagnostic. One resolver would receive all reports and executor traces. M4 assigns one specification analyst, one candidate critic, and one oracle critic. The analyst sees no program. Each critic sees the specification and only the assigned program. A deterministic merge forms a provisional decision before one resolver sees both programs, all reports, and executor evidence.

Both arms specify the same requested model, three parallel upstream calls, one resolver, and the same deterministic executor contract. Each upstream role may propose at most three evidence slots and two executable claims. The executor applies no more than six claims per arm to both programs and records returns, exceptions, timeouts, and observation differences. Proposed expected values carry no truth status. Actual token use and latency would be measured rather than assumed equal.

The planned contrast bundles role specialization, view restriction, and evidence routing. It cannot isolate any one mechanism. No G3R or M4 main call was made, so the study supports no comparison between them.

4.3 Mirror analysis unit and original resource targets

Let C_A be one qualified clean program. Let D_B be a qualified fault derived from a second clean lineage C_B . One mirror candidate contains

$$I_1 = C_A \mapsto \text{Candidate}, \quad D_B \mapsto \text{Oracle}, \quad (9)$$

$$I_2 = D_B \mapsto \text{Candidate}, \quad C_A \mapsto \text{Oracle}. \quad (10)$$

The clean program paired with D_B is not its immediate parent. Mirror mates occupy nonadjacent contexts. Fault role and code-block order are balanced separately. Both-clean, both-faulty, and ambiguity items occupy task-disjoint supplemental strata.

For arm a and task t , $Y_{a,t} = 1$ only when both presentations receive the exact gold label. The planned paired risk difference is

$$\Delta = \frac{1}{T} \sum_{t=1}^T (Y_{M4,t} - Y_{G3R,t}). \quad (11)$$

The locked analysis specifies an exact two-sided McNemar test at $\alpha = 0.05$ [14]. Power integrates the exact conditional rejection rule over the binomial discordance count [9]. The stated alternative uses $\delta = 0.2$ and discordance $d = 0.5$. The helper parameterizes discordant cells as $p_{10} = (d + \delta)/2$ and $p_{01} = (d - \delta)/2$, with $|\delta| \leq d$. Historical protocol records name 60 mirror tasks as the original resource target and 40 as the minimum resource tier. The original target receipt remains unchanged.

A pipeline receipt links construction, freeze, qualification, witness generation, program review, independent testing, promotion, and selection to verified predecessors. The anonymous artifact preserves those records and the protocol materials.¹

5 Executed Pipeline and Attrition

No quantity in this section comes from a G3R versus M4 main run. Execution ends after label-free selection. Table 1 reports the completed gates.

All 6,674 qualification jobs completed on the first attempt. The 39 clean-suite disagreements were machine-suite outcomes. Of 692 mutation base survivors, 58 were excluded because their clean parent did not pass both machine suites. The witness stage ran 634 suites. Of these, 217 yielded a returned-output witness. The fixed per-parent review cap retained 81 fault candidates without consulting human verdicts.

Primary reviewers agreed on 275 of 277 packet verdicts. Both disagreements concerned separately constructed implementations of the same median task. The prose required the median while a doctest expected 15.0 for a list whose standard median is 8.0. The resolver classified both packets as materially ambiguous. This result shows why the label space cannot force a choice between candidate and oracle fault.

The independent stage scheduled 180 programs with correct panel verdicts and two distinct fresh inputs. One program disagreed with the designated reference. For HumanEval/89 and input Az-9_q, the program returned Ed-9_u while the designated reference returned Ad-9_u. The protocol retained this single-case non-pass for investigation. The disagreement does not establish that the designated reference is correct.

Promotion accounted for every reviewed program. The 277 decisions yielded 170 qualified programs and 107 explicit attritions. The attritions comprised 18 review-plan exclusions, 88 duplicate-source exclusions, and the independent non-pass. The plan exclusions comprised 13 packets without distinct independent inputs, four material-ambiguity verdicts, and one clean candidate judged faulty.

A returned-output witness also exposed a boundary defect. On HumanEval/81, a grade value of 1.3 produced C- where the specification required D+. Both primary reviewers classified the mutated program as faulty.

The 101 clean and 69 fault programs did not form enough task structures. The resource ceiling in Table 2 ignores task identity and

¹<https://anonymous.4open.science/r/masgain2026-artifact>

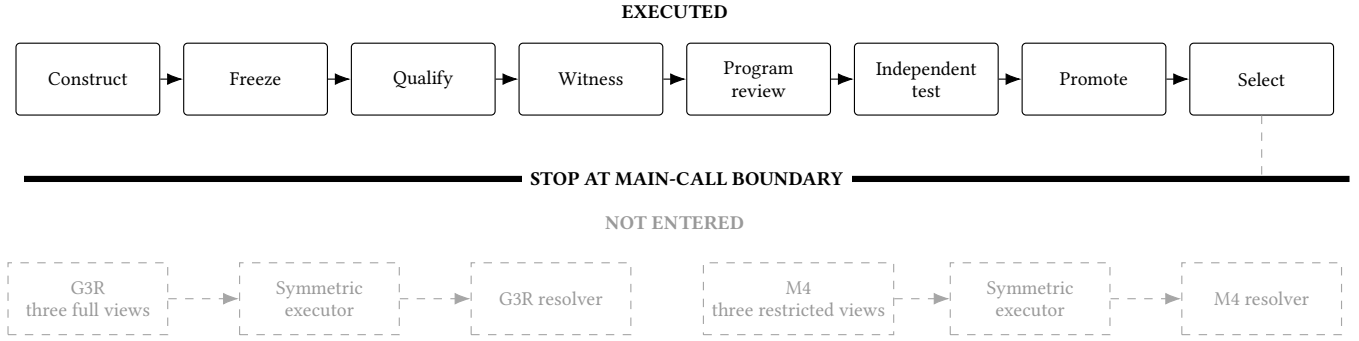


Figure 1: Executed preparation and the unentered comparison. Solid black gates produced verified receipts. Dashed gray protocol stages were never entered. The bold bar marks the preserved no-call boundary.

Table 1: Executed preparation flow and stopped comparison boundary

Stage	Locked input	Executed units	Recorded disposition
Construction	114 tasks	228 jobs and 230 physical attempts	228 accepted source programs with one infrastructure retry and one quota pause
Mutation freeze	228 parents	6,218 generated mutations and zero executions	Outcome-free ranked universe
Qualification	456 clean suites and 6,218 mutation suites	6,674 terminal jobs	417 clean-suite agreements, 39 clean-suite disagreements, 692 mutation survivors, and 5,526 detections
Witness	692 survivors	634 terminal suites	58 were excluded. 217 suites yielded a returned-output witness and 417 did not
Program review	196 machine-clean programs and 81 fault candidates	277 double-reviewed neutral packets	275 primary verdict agreements and two resolved disagreements
Independent test	180 clean-program suites	180 jobs with two fresh cases each	179 agreements and one disagreement
Promotion	Verified stage lineage	277 reviewed programs	101 clean and 69 fault programs qualified. Attrition included 18 plan exclusions, 88 duplicate-source exclusions, and one independent non-pass
Selection and boundary	Verified promoted pool	51 selected items and eight pre-gold mirror-task candidates	Zero final-gold records and zero of 408 planned main model calls

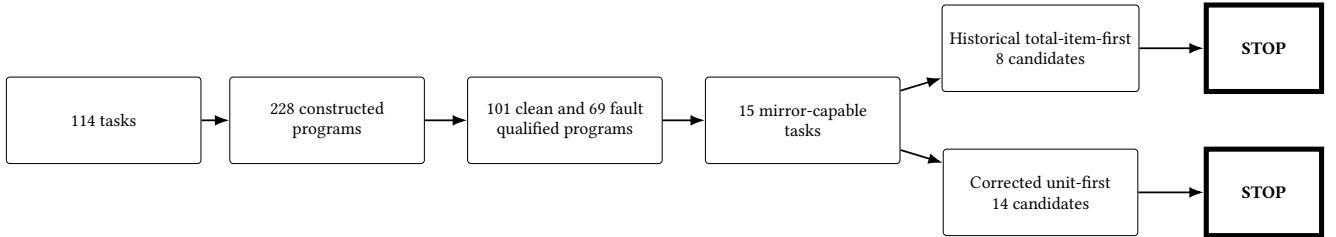


Figure 2: Compound-unit attrition. Program supply narrowed under task identity and assembly constraints. The executed historical allocation contained eight candidates. The later unit-first re-audit found 14 under the same pilot quotas. Both pre-gold allocations remained inadequate before final adjudication.

therefore overstates the usable supply. The identity ceiling requires two clean lineages and a qualifying fault within one task.

The resource ceiling is 50. Task identity lowers that ceiling to 15. The executed historical search maximized total items and selected 51 items containing eight pre-gold mirror-task candidates. This objective required a separate audit because maximizing all items does not establish the largest feasible mirror count.

6 Method Validation

6.1 Preserved unit-accounting correction

A retrospective audit of the preserved receipt trail found a denominator error while the prospective no-call boundary remained closed. One superseded selection receipt supplied the 51-item design-wide count to the paired power calculation. The primary analysis operates on mirror tasks, not all selected items. The correction in Table 3 is a direct validation of typed analysis-unit accounting.

Table 2: Attrition from promoted programs to candidate primary units

Stage	Program or item supply	Primary-unit status	Retention/ conversion factor
Promotion pool	101 clean and 69 fault programs	Primary units not yet assembled	Reference supply
Resource ceiling	Same promoted pool	At most 50 pairs from program counts alone	$\min(\lfloor 101/2 \rfloor, 69) = 50$
Identity ceiling	Same promoted pool	15 mirror-capable tasks	$15/50 = 0.30$
Historical total-item-first selection	51 main items with 16 mirror presentations	8 pre-gold mirror candidates in the historical allocation	$8/15 = 0.53$
Corrected unit-first selection under pilot quotas	63 potential main items with 28 mirror presentations	14 pre-gold mirror candidates in the re-audit	$14/15 = 0.93$
Original resource tiers	Historical allocation of eight candidate pairs	Insufficient for both historical tiers	$8/40 = 0.20$ and $8/60 = 0.13$

Table 3: Preserved unit-accounting validation case

Receipt interpretation	N	Decision
Superseded all selected items	51	Invalid
Corrected mirror tasks	8	STOP

Table 4: Lexicographic selector re-audit

Constraint case	Mirror	Both-clean	Both-faulty	Main items
No supplemental minima	15	26	0	64
One per supplemental stratum	14	26	1	63
Historical quotas of 26 and 1	14	26	1	63

Ten development identities remain excluded. Eight challenge items from a ten-identity reservation pool remain fixed in every case.

The preserved receipts did not autonomously detect the denominator error. They retained the unit identities, predecessors, and assumptions needed for a later audit to detect it, reproduce its consequence, and issue a successor record without erasing the historical one.

6.2 Selector objective audit

The pilot selector maximized total selected items rather than primary mirror tasks. We therefore reran allocation with Eq. 2 and no model calls. The audit considered the mirror ceiling without supplemental constraints, the ceiling with one item required in every supplemental stratum, and the ceiling under the historical pilot quotas. Table 4 reports the resulting allocations. Fixed development and challenge reservations remain unchanged in all three cases.

The executed count of eight was not primary-unit-maximal. It resulted from a sequential objective that maximized total items first. The lexicographic audit found 14 mirror tasks under the historical pilot quotas and the identity ceiling of 15 when no supplemental minimum applied. The original target supplemental quotas of ten both-clean and ten both-faulty items were infeasible because only nine tasks were both-faulty-capable. The minimum supplemental quotas of eight and eight supported at most seven mirrors.

This correction does not reopen the study. Even the counterfactual pre-gold identity ceiling of 15 mirror tasks has power 0.098 under the locked analysis. The audit matters because a claim of maximality must follow the objective for the primary unit.

Table 5: Power under the pre-specified alternative

Mirror pairs	Rounded exact-test power
8	0.016
15	0.098
40	0.354
60	0.525
103	0.802

6.3 Independent adequacy audit

The repository statistics helper recomputed unconditional exact two-sided McNemar power under $\delta = 0.2$, $d = 0.5$, and $\alpha = 0.05$. Table 5 reports rounded paper values. Exact floating-point values remain in the successor receipt and artifact. A separately implemented exact enumeration, written without importing the repository helper, reproduced all reported values. The artifact preserves both implementations and their hashes.

The separate implementation directly enumerates the trinomial task outcomes, while the repository helper conditions on the number of discordant pairs. Both reproduced power of 0.0158 for eight historical candidates and found that the original 60-pair resource target would provide only 0.525 power under the stated alternative. We preserve the original target receipt and record this correction in a successor adequacy receipt. The first integer sample size that reaches 0.8 power is 103 pairs.

The correction also calibrates the historical tiers. At target power 0.8, the minimum detectable risk difference is about 0.317 with 40 pairs and 0.264 with 60 pairs. Neither tier supports the pre-specified effect of 0.2 at target power. No feasible risk difference under the assumed discordance could attain 0.8 power with eight task-level units. The discordance-bound maximum is 0.145.

The observed supply chain was therefore 50 pairs under a resource-only count, 15 mirror-capable tasks under identity, and eight mirror-task candidates under the executed selection. The re-audit found 14 under the historical pilot quotas. Final adjudication could only decrease either candidate count. Even the identity ceiling remained inadequate, so the adequacy function returned STOP before final gold and before every planned main call.

7 Discussion and Limitations

7.1 Compound attrition as a design parameter

The relevant yield in this study was not programs per task. It was complete evidence structures per eligible primary analysis unit. A

mirror needed two accepted clean lineages, one independently witnessed fault, a valid role swap, and later final adjudication. Both faulty controls needed faults under distinct parents. Review and fresh-input requirements removed programs that had passed machine suites. These dependencies made task-level yield far lower than program-level yield.

Future agentic studies should estimate each conversion before fixing a confirmatory design. The sequence is to declare the analysis unit aligned to the estimand, record program-level retention, assemble task structures under all reservations, and compute power from the eligible compound units. A small construction pilot can estimate these rates without entering the main comparison.

The adequacy correction adds a second lesson. A resource target is not a power target until the stated test, effect, discordance, and sample size reproduce the desired power. A successor receipt can repair the operational plan while retaining the historical record needed to audit the change.

7.2 Scope and interpretation

The executed stages establish an inspectable path from construction through selection. They support claims about gate behavior, attrition, accounting, and the stop boundary. They do not support claims about G3R, M4, routed evidence, specialist roles, generalist review, or comparative protocol performance.

The construction universe contains short Python tasks from HumanEval+ and a bounded mutation family. Qualification yield may differ for production code, another language, or naturally occurring defects. Two lineages came from isolated sessions with distinct prompts, but they shared a model family and style contract. Their errors may remain correlated.

Human program review is fallible. Neutral packets, independent answers, fresh inputs, and resolution make decisions inspectable. Residual judgment error remains possible. A disagreement with the designated reference remains a prompt for investigation until the specification settles correctness.

The exact power calculations depend on the pre-specified risk difference and discordance. They diagnose the recorded design under its own assumptions. They do not establish power under every alternative analysis. The selector audit is also conditional on the frozen promoted pool, eligibility rules, reservations, and quota definitions.

The planned treatment changes role specialization, view partitioning, and evidence routing together. Even an adequately powered future run would not isolate those mechanisms. Token use, latency, abstention, and system failures remain unmeasured because the main protocols were not entered.

Original raw model streams and human administrative records remain outside the anonymous repository. This limits third-party replay of the private evidence chain. The evidence root remains local and under operator control. A motivated operator could delete or replace it. The receipts therefore provide no tamper evidence against that operator, and the study used no external anchor. Content hashes establish equality to preserved bytes while study judgment remains a separate question.

8 Conclusion

Receipt-Gated Compound-Unit Accounting makes compound-unit attrition explicit before a confirmatory comparison begins. In this case, 101 qualified clean programs and 69 qualified fault programs yielded 15 mirror-capable tasks. The historical total-item-first allocation held eight pre-gold mirror candidates, and the corrected unit-first re-audit found 14 under the same pilot quotas. Even the 15-task identity ceiling remained far below adequacy. The receipt trail preserved the unit-accounting error and enabled an audit to detect and correct it before protected outputs existed. The same retrospective audit found that the original 60-pair resource target supplied only 0.525 power under the stated alternative. The prospective no-call boundary kept every planned main comparison call closed.

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